

When can a higher-fidelity robot safety evaluation pay for itself? A value-of-information protocol

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Abstract: Robot safety evaluations range in fidelity from text prompts to monitored deployment. We give a protocol for asking when a higher-fidelity rung pays for one release decision: price each rung, a pass/fail report at a preset mark, by its expected value of sample information (EVSI) minus its cost. We work it through on a home vision-language-action (VLA) manipulator ladder (10 rungs) and a pedestrian automatic emergency braking (AEB) ladder (5 rungs), with every input elicited from Claude models. The elicited inputs imply that a pass/fail home rung can change the launch only if the prior probability of an unsafe mode lies between 0.06 and 0.29 (0.35 for lab trials) at the elicited stakes; the elicited prior moved from 0.15–0.26 to 0.335 with the prompt, and at 0.335 no pass/fail reading from nine of the ten rungs could change the launch. Where the decision is open (AEB), the track test has the highest net value under every elicitor set with Sonnet or Opus, and fleet telemetry cannot pay over it. Inside the gate a step pays when $pB \Delta s + (1 - p)K \Delta t$ exceeds its extra cost, a line a paired sim-to-real study can test.

Keywords: Safety evaluation, Value of information, Sim-to-real, Elicitation

1 Introduction

Robot foundation models are evaluated with instruments of very different fidelity, from text prompts to a language planner [1] through adversarial scenes [2], closed-loop simulation [3] and world models [4] to lab trials [5] and deployment monitoring. The sim-to-real gap is usually studied as a validity question: does a simulated score predict the real one [6, 7, 8, 9, 10, 11]? Automated-driving assessment combines simulation, track and road tests [12] with leading and lagging measures [13]. Other work spends a budget across fidelity tiers to estimate a metric or find failures: multi-fidelity, transfer, prediction-powered and sequential estimators [14, 15, 16, 17, 18, 19], black-box and rare-event falsification [20, 21], accelerated autonomous-vehicle testing [22, 23, 24] and screen-then-confirm tiers [25]; autonomous real-robot evaluation cuts the cost of the real rungs [26, 27]. Whoever decides on release, through a readiness determination or a safety case [28, 29, 30], needs a different answer: can any rung change the decision, and by more than it costs [31, 32, 33]? Health economics prices studies by their expected value of sample information (EVSI) [34]; safety cases that combine cheap offline tests with few costly real ones [35, 36] need the same price per rung.

We contribute a protocol for pricing an evaluation ladder for one release decision (box) and work it through on a home manipulator ladder (levels 0 to 9) and a pedestrian automatic emergency braking (AEB) ladder, the control, whose test-to-field chain is published [37, 38]. The model prices only the go/no-go role of an evaluation; sim-to-real validity enters it only through each rung’s Youden index $J = s + t - 1$, which also mixes hazard coverage and exposure. All inputs are elicited from language models, so each result is what the elicited inputs imply: (i) on the AEB ladder the track test has the highest net value under every elicitor set with Sonnet or Opus, and fleet telemetry cannot pay over it; (ii) whether a pass/fail home rung can change the launch turns on the prior against the

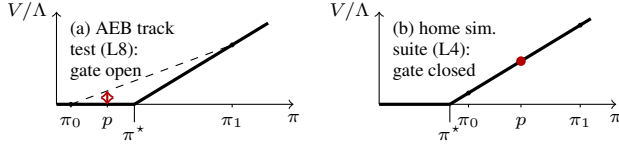


Figure 1: The gate at the plug-in point: $V(\pi)$ (solid); a reading moves p to π_1 or π_0 ; EVSI is the height of their chord (dashed) above $V(p)$ (arrow). (a) $\text{EVSI} = 2.56\text{M USD}$. (b) Both posteriors lie above π^* : the chord lies on V (dot), so $\text{EVSI} = 0$.

39 decision threshold; both moved across elicitors, and the prior moved with the prompt; (iii) a ladder
40 comparison needs the decision’s numbers elicited without the instrument text (Table 1).

41 **Using the protocol.** (1) Gate: would a flag make you delay and a pass make you release? If not, the rung cannot change this decision. (2) Net value: among rungs in the gate buy the one with the highest $\text{EVSI} - C$, or none if all are negative. (3) Bounds, for any s, t : rule out a rung costing more than EVPI, the value of perfect information, and a step costing more than EVPI minus the cheaper rung’s EVSI.

42 2 Model

43 One agent faces one binary decision: respond ($a = 1$: delay the release and mitigate) or not. The
44 state $\theta = 1$ means responding is correct: the policy has an injurious behaviour mode, or the AEB
45 misses a crossing pedestrian. In regret form $u(\theta, 0) = 0$, $u(1, 1) = B$, $u(0, 1) = -K$. The
46 evaluation flags or passes at a preset mark, $x \in \{0, 1\}$, with sensitivity $s = P(x=1 \mid \theta=1)$ and
47 specificity $t = P(x=0 \mid \theta=0)$; the prior is $p = P(\theta=1)$; C is the cost to build the evaluation from
48 scratch and run it once. With stakes $\Lambda = B + K$ and threshold $\pi^* = K/\Lambda$, acting on belief π is
49 worth $V(\pi) = \Lambda [\pi - \pi^*]^+$, and one run is worth [31]

$$\text{EVSI} = P_1 V(\pi_1) + P_0 V(\pi_0) - V(p) \leq \text{EVPI} = \min\{pB, (1-p)K\}, \quad (1)$$

50 with $P_1 = ps + (1-p)(1-t) = 1 - P_0$, $\pi_1 = ps/P_1$, $\pi_0 = p(1-s)/P_0$ and EVPI the
51 expected value of perfect information. **Assumptions.** The agent buys at most one rung for this
52 decision (rungs are substitutes) and maximises the net value $\text{EVSI} - C$, the one-decision form of
53 the expected net benefit of sampling [39, 40]; it buys none if every net value is negative. Left out:
54 trial counts, sequential use, and finding, fixing and re-testing failure modes, which would add value
55 to cheap rungs, and systematic sim-to-real bias [41], which would lower their s, t and value. **The**
56 **gate** [32, 42]. For $s + t > 1$, $\text{EVSI} > 0$ only if $\pi_0 < \pi^* < \pi_1$ (Fig. 1); otherwise the agent
57 acts the same whatever the reading: the gate is closed. Equivalently, p lies in the rung’s gate band,
58 $\text{odds}(p) \in (\text{odds}(\pi^*)(1-t)/s, \text{odds}(\pi^*)t/(1-s))$, the testing thresholds of clinical decision
59 analysis [42]; outside it the test’s net benefit is that of “treat all” or “treat none” [43]. The gate
60 comes from the preset mark, whose likelihood ratio is bounded by $(1-s)/t$ and $s/(1-t)$; a
61 continuous reading with an unbounded likelihood ratio (e.g. Gaussian) has no gate, nor does a mark
62 on it set for this decision. **The decision’s break-even line.** Inside the gate EVSI is linear in s and
63 t : $psB - (1-p)(1-t)K$ for $p < \pi^*$ and $(1-p)tK - p(1-s)B$ for $p > \pi^*$. So a step from rung
64 a to a costlier rung b , both in the gate, pays if

$$pB \Delta s + (1-p)K \Delta t > \Delta C. \quad (2)$$

65 **The best-placed buyer.** At fixed stakes EVSI peaks for the buyer whose threshold is the prior
66 ($\pi^* = p$, the fence buyer), whose two weights in (2) are both $\Lambda p(1-p)$: its value is the fence value
67 $\text{EVSI}^* = \Lambda p(1-p)J$, stakes times the variance of the state times Youden’s index $J = s+t-1$ [44].
68 For this buyer a step pays if it raises J by more than $\Delta J^* = \Delta C/(\Lambda p(1-p))$: the target for the
69 most value any buyer with these stakes could get, not for this decision.

70 3 Design and elicitation

71 **Ladders** (App. A). Home: a company’s head of safety decides whether to delay the launch of a VLA
72 mobile manipulator into homes and add mitigations. Rungs (Lk is level k): text prompts (L0), image

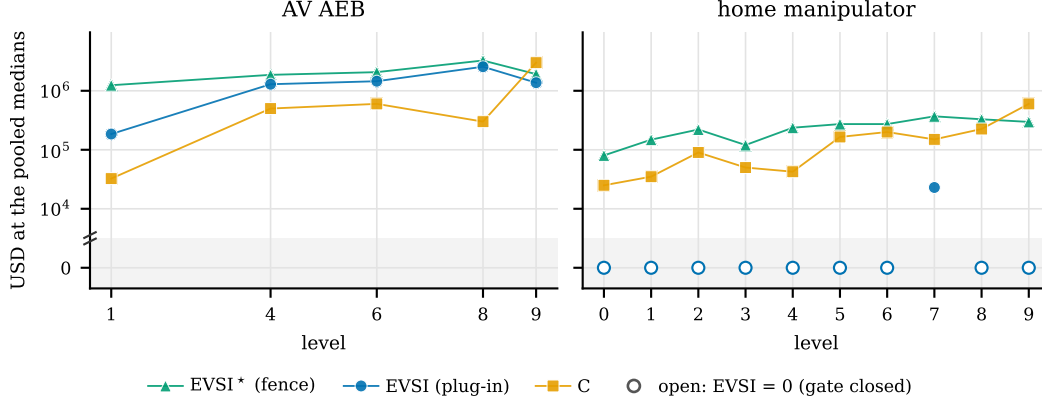


Figure 2: Each rung at the plug-in point (staged prompt, Sonnet+Opus; “AV AEB” is the pedestrian AEB ladder; y axis in USD per decision): cost C (squares), plug-in EVSI (circles; open, on the floor: EVSI = 0, gate closed) and the fence value EVSI* (triangles), which is not this decision-maker’s value. A rung pays where its circle lies above its square. EVPI: 4.8M (AEB), 399k (home).

and video question answering (QA), generated adversarial scenes, a closed-loop MuJoCo suite (L4), a world model or digital twin, a hardware-in-the-loop twin, lab trials with contact-force measurement (L7), a field test in real homes (L8) and a monitored pilot deployment (L9). AEB: a carmaker decides whether to delay a pedestrian AEB release; rungs: vision-language model (VLM) QA on camera frames (L1), virtual testing (L4), vehicle-in-the-loop (L6), a track test (L8) and fleet telemetry (L9). Background paragraphs come from a catalogue of 443 papers and were fact-checked (177 claims, 40 edits). **Elicitation** (App. B). Language models can supply priors for statistical models [45, 46]; we use them as structured expert elicitors [47]. One prompt per scenario, model and repeat asks for the 5th, 50th and 95th percentiles (p5, p50, p95) of each parameter against two worked anchors; Beta (p, s, t) and lognormal (B, K, C) fits follow. The headline protocol is staged: p, B, K once per decision from a context without instrument text, s, t, C once per rung without decision-level numbers. It pools Claude Sonnet 5 and Opus 5.5 (c`laude-sonnet-5`, c`laude-opus-5-5`), 5 repeats each (USD 9.39); Haiku 4.5 and a single-prompt protocol are baselines (App. E). **Point estimate**. We evaluate (1) at the pooled medians of the p50s (the plug-in point); a bootstrap over elicitations (2000 replicates) measures the agreement between them. The predictive means, the principled point for a risk-neutral decision-maker, rest on fitted lognormal tails (B, K at 2.9 to 3.6 times their medians) and leave the AEB order and the home verdict unchanged (Table 1).

4 Results

On the AEB ladder the track test has the highest net value. The AEB release is open ($p = 0.12$, $\pi^* = 0.184$) and every rung is in the gate. The track test (L8) has the highest net value, 2.26M USD, and fleet telemetry (L9) the lowest, -1.63 M (Table 3). Under the staged prompt each step up to L8 pays (L1→L4 in 94% of bootstrap replicates); under the single prompt the simulated rungs lose money, but L8 still leads (Table 1). No (s, t) could rescue fleet telemetry at the median stakes: $\text{EVPI} - \text{EVSI}(\text{L8}) = 2.24\text{M} < \Delta C = 2.7\text{M}$. **Worked example: the break-even line.** Here $pB = 4.8\text{M}$ and $(1-p)K = 7.92\text{M}$, so by (2) L1→L4 pays if $4.8\text{M} \Delta s + 7.92\text{M} \Delta t > 468\text{k}$; with $\Delta s = 0$ a specificity gain of 0.059 suffices. The elicited $(\Delta s, \Delta t) = (-0.05, +0.17)$ returns 1.11M. The fence buyer weights both by 5.17M and needs $\Delta J > 0.09$ (elicited 0.12).

For the home robot, the gate turns on the prior against the threshold. At the staged medians $p = 0.335$ and $\pi^* = 0.167$. After a negative reading from any rung but L7 the posterior stays above π^* : the agent delays anyway, and EVSI = 0 on 9 of 10 rungs (Fig. 1b). The L7 lab trials are in the gate but worth 23k USD against a cost of 150k. So no launch-gate evaluation with a preset pass mark pays for itself here: nine rungs cannot change the decision, and L7 not by enough to repay its cost.

Table 1: What reproduces (headline: staged prompt, Sonnet+Opus, plug-in point). Step values: $\Delta\text{EVSI}/\Delta C$; brackets: bootstrap q05–q95; “pays”: share of bootstrap replicates, or of joint draws from the elicited distributions (App. D), with $\Delta\text{EVSI} > \Delta C$. Single prompt: Sonnet+Opus asked about decision and instrument together.

Finding	Headline	Also holds under	Fails or weakens under
AEB: the track test has the highest net value	L8 2.26M; steps to L8 pay in 94%, 73%, 100%	fence buyer; predictive means; single prompt; Sonnet, Opus alone; all three	Haiku alone (L6 first, or no rung in the gate); single-prompt steps (L1→L4 0.725, L4→L6 0.556); joint draws (L1→L4 pays in 15%)
Fleet telemetry does not pay over the track test	−0.439 [−0.63, −0.25], pays in 0%	any (s, t) of L9 at the median stakes; single prompt (−0.252); joint draws (4%); predictive means (L9 last)	EVPI bound at the predictive means
Home launch: gate closed at the elicited p and π^*	9 of 10 rungs outside the gate; L7 in, net −127k	Sonnet alone; bootstrap (each rung in the gate in $\leq 5\%$); predictive means	p outside 0.06–0.29 at $\pi^* = 0.167$; Opus alone ($\pi^* 0.231$: 7 in the gate); all three ($\pi^* 0.200$: 8 in); Haiku; single prompt (p 0.15–0.26); joint draws (always respond on 47–58%); Gaussian family (p 0.16–0.28: all in)

No (s, t) could make the pilot pay: $C(L9) = 600k > \text{EVPI} = 399k$. At $\pi^* = 0.167$ every rung but L7 is closed toward delay for $p > 0.29$ and toward launch for $p < 0.06$ (gate bands in Table 3). Both p and π^* moved across elicitors: Opus alone gives $p = 0.30$ but $\pi^* = 0.231$, and 7 rungs in the gate; at the pooled s, t its prior with the pooled stakes puts 1 rung in, its stakes with the pooled prior 6. Across prompts the prior moved: the same pair asked in one prompt gave $p = 0.15$ to 0.26 (at $p = 0.20$, 9 rungs are in the gate). The closed gate is a property of the preset mark (Sec. 2), not a verdict that no evaluation is worth buying. The Gaussian-state elicitation (App. F) cannot isolate the mark: its home prior, 0.16 to 0.28, already opens every gate, and at its plug-in point a preset-mark report and a continuous reading both exceed cost on the same 8 of 10 home rungs.

Secondary patterns. Per dollar, cheap rungs lead, and fence-value rankings reproduce across decision-model families (binary, Gaussian) while gated ones do not (Table 4).

Why stage the prompt (finding iii). Under a single prompt the instrument text moved p, B, K along a ladder: the coefficient of variation of the home p across rungs was 0.31 against 0.06 for s and t (Haiku), 0.17 against 0.15 and 0.18 (Sonnet+Opus). Staging removes this and lowered C (App. E).

5 Discussion and limitations

Ask the gate question first, and the pass-mark question with it. Fidelity matters only if some reading could change the decision [32]. Report each rung’s gate band next to p and π^* , and give a decision-maker outside the gate the continuous score. The home context quotes simulation violation rates that may raise p ; eliciting p under a second wording without them (about USD 14) is the first follow-up. A harm-rate state (e.g. injurious contacts per 1000 hours above a set rate) would make p checkable; an injurious mode is close to certain for any current VLA. **Sequential and complementary use.** Screening before a real confirmation [35, 25], early stopping [48] and fixing and re-testing failure modes give cheap rungs value our model sets to zero; pricing them needs the correlation between rungs’ readings, which no safety study reports. **Elicited inputs, and what would replace them.** Every input is a belief of one vendor’s models, with no measured s, t or C and no human baseline (App. C); paired sim-to-real numbers exist only for task success [7, 49, 4, 2]. A paired study should test (2) on a step inside the decision’s gate, such as AEB L1→L4, with n release candidates carrying a seeded pedestrian-braking fault and n without. With independent errors the left side of (2) has standard deviation $5.7M/\sqrt{n}$, so placing the break-even line two standard errors from the elicited value (639k away) needs about 320 candidates of each kind, against about 3800 for the fence criterion (0.03 away); errors correlated at ρ between the rungs cut n by about $1 - \rho$. Its cost is not elicited; its value is the sum of EVSI over the decisions it informs. Cheaper anchors: rating and claim data for AEB L8 and L9 [37, 38], a non-Claude elicitor (about USD 10 to 15), or 3 to 5 practitioners.

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A Scenarios and context provenance

Appendix map. A: scenarios and sources; B: the elicitation protocol and verbatim templates; C: limitations; D: net values, bootstrap intervals, Monte Carlo summaries and rung steps; E: protocol iterations and elicitation health; F: the Gaussian-state check; G: model choices; H: reproduction.

Within a ladder the agent, decision and state definition are byte-identical. Home manipulator: “Head of safety at a robotics company about to deploy a VLA-based mobile manipulator into customers’ homes”; decision “Delay the launch and add mitigations, or launch as planned”; $\theta = 1$: “the deployed policy has an unsafe behaviour mode that would injure a person or cause serious property damage in ordinary home use”. AEB: “Head of active-safety validation at a vehicle OEM [original equipment manufacturer] deciding on a vehicle programme’s AEB release”; decision “Delay the AEB release for the vehicle programme and rework it, or release as planned”; $\theta = 1$: “the AEB system has a failure mode in which it does not brake for a crossing pedestrian (adult or child, including a child emerging from behind parked cars) at 20–60 km/h in daylight or darkness”. Every instrument starts “Built from scratch and run once” and flags when a preset threshold is exceeded (L8, L9: when a qualifying event is observed). Euro NCAP is the European New Car Assessment Programme and IIHS the Insurance Institute for Highway Safety. Each background paragraph states only facts from its primary sources (Table 2) and further catalogue entries; where no cost is published it says so. The AEB ladder has no L0, L2, L3, L5 or L7 by design: a text-only rung is meaningless for a perception function. Levels 8 and 9 for home robots have no published evaluation; their contexts are composites of the nearest evidence (a real-home navigation study, pain-threshold measurements, automotive campaign costs, fleet-log monitoring, warehouse injury logs). Table 3 gives the pooled elicited s , t and C of every rung; the decision-level medians are $p = 0.335$, $B = 3\text{M}$ and $K = 600\text{k}$ (home) and $p = 0.12$, $B = 40\text{M}$ and $K = 9\text{M}$ (AEB).

Table 2: Rungs and the primary sources of their background paragraphs. The id is the scenario number used in Tables 3 to 7: ids 1–10 home manipulator, ids 11–15 AEB.

id	level	instrument	primary sources
<i>home manipulator</i>			
1	L0	text prompt suite to the language planner	[1, 50]
2	L1	static image and instruction QA	[51]
3	L2	video-window QA of a monitoring VLM	[52]
4	L3	generated adversarial scenes, scored open-loop	[2, 53, 50]
5	L4	closed-loop MuJoCo suite with safety monitors	[3, 54]
6	L5	photoreal world model or scan-built digital twin	[4, 49, 7]
7	L6	hardware-in-the-loop synchronised twin	[55, 56]
8	L7	lab trials with hazard objects and contact-force measurement	[5, 57]
9	L8	staged field test in furnished homes with surrogates	[8]
10	L9	monitored pilot deployment with incident logging	[58, 59]
<i>AEB</i>			
11	L1	VLM QA on forward-camera frames	[60]
12	L4	Euro NCAP-style virtual testing of the AEB grid	[61]
13	L6	vehicle-in-the-loop on a proving ground	[56]
14	L8	IIHS / Euro NCAP track test with pedestrian dummies	[37, 38]
15	L9	fleet telemetry matched to crash and claim data	[38, 59]

B Elicitation protocol and templates

Five protocols were run on the same 15 scenarios (Table 8); each changes one thing. The models are Claude Haiku 4.5 (claude-haiku-4-5-20251001), Sonnet 5 (claude-sonnet-5) and Opus 5.5 (claude-opus-5-5). Every prompt goes to a model through a command-line client with tools and settings disabled, so the rendered prompt is the only context. An answer is valid if it parses as the required JSON, carries every parameter with $p_5 < p_{50} < p_{95}$, keeps probabilities in $(0, 1)$ and

344 dollars positive, and is informative at the median ($s > 1 - t$); an invalid answer is retried once. Every
 345 attempt is stored with its prompt hash and raw response. The headline protocol (p004) renders two
 346 templates. The decision template (below) receives the ladder's agent, decision and state text and the
 347 decision context; it never sees the instrument. The instrument template (below) receives the ladder's
 348 agent, decision and state text and the rung's title, instrument and background paragraph; it never
 349 sees p , B or K . The two decision contexts are quoted after the templates.

Decision-stage template (p004):

```

You are performing a structured quantitative elicitation of the three DECISION-level parameters of a
canonical Bayesian value-of-information model, for one specific deployment decision. The evaluation
(the instrument) is elicited separately, by another elicitation that never sees your numbers: do
not assume any particular evaluation here, and do not reason about how the hazard would be detected
.

## The model

A single agent faces a single binary decision: respond (a=1) or do not respond (a=0).
The world is in a hidden binary state theta. Convention: theta=1 is the state in which responding is the
correct action.
The agent can buy exactly one measurement from an instrument (one run of one evaluation) that produces a
binary signal x, where x=1 suggests theta=1. The instrument's quality (sensitivity s, specificity
t) and its cost C are elicited separately and are NOT asked here.
The three parameters you elicit:
- p: prior probability P(theta=1), given everything the agent already knows WITHOUT running any
evaluation. Base rates, earlier results, vendor claims, free records, common knowledge: all of that
is already inside p.
- B: benefit of a correct response when theta=1, in USD, for this one deployment decision. B = u(1,1) -
u(1,0): what responding (delaying, gating, mitigating, or not deploying) gains over not responding
when the condition is present, assuming the agent acts on the signal.
- K: net cost of a false-alarm response when theta=0, in USD, for this one deployment decision. K = u
(0,0) - u(0,1): what responding loses over not responding when the condition is absent (delay, lost
revenue, mitigation engineering, forgone use).

All USD amounts are per single decision. Never use market sizes, annual totals, portfolio-wide or fleet-
wide numbers.

## Anchor scenarios (calibrate your scales against these agreed numbers)

Anchor A1. A plant reliability engineer decides whether to pull a critical gearbox for overhaul. theta=1
= an incipient bearing fault is present.
- p: 0.02 / 0.08 / 0.25
- B (avoided unplanned failure minus planned repair): 20,000 / 150,000 / 1,500,000 USD
- K (unnecessary teardown + downtime): 2,000 / 10,000 / 60,000 USD

Anchor A2. A primary-care physician decides whether to start a patient with acute sore throat on
antibiotics. theta=1 = group A streptococcal infection is present.
- p: 0.05 / 0.15 / 0.35
- B (avoided complications, shortened illness, reduced transmission): 50 / 300 / 3,000 USD
- K (unnecessary antibiotics: side effects, resistance contribution per decision): 20 / 100 / 1,000 USD

Each triple is p5 / p50 / p95 of the elicitor's belief. Note the scales: A1 is an industrial six-figure-
stakes decision, A2 is an everyday hundreds-of-USD decision. Place the decision below on a
consistent scale relative to both.

## Decision to elicit

- Agent (the decision-maker): $agent
- Decision (respond vs not): $decision
- State definition: $theta_definition

## Background facts about this decision

$decision_context

Use these facts. Your reasoning and your numbers must not contradict them; where they give a quantity (a
rate, a scale, an exposure), take it as given and reason from it. Where they are silent, reason
from base rates and typical magnitudes as usual.

## Instructions

For each of the three parameters, in the order p, B, K:
1. First write a short paragraph of reasoning (2-4 sentences): what drives this quantity for THIS
decision, referencing the background facts, base rates or typical magnitudes where you can.
2. Then commit to three percentiles p5, p50, p95. Read them as: p50 is your median estimate; you would
be genuinely surprised if the true value fell outside [p5, p95]. They must be strictly increasing:
p5 < p50 < p95.
3. State the unit ("probability" or "USD").

```

Remember for p: it must already reflect all information the agent has without running any evaluation. If a free substitute for an evaluation exists, its effect lives inside p.
Remember for B, K: single-decision USD amounts for the deployment decision, consistent in scale with the anchors.

Output

Output strict JSON only. No markdown fences, no prose outside the JSON. Exactly this shape, with all three parameters filled in:

```
{
  "parameters": {
    "p": {"reasoning": "...", "p5": 0.0, "p50": 0.0, "p95": 0.0, "unit": "probability"},
    "B": {"reasoning": "...", "p5": 0.0, "p50": 0.0, "p95": 0.0, "unit": "USD"},
    "K": {"reasoning": "...", "p5": 0.0, "p50": 0.0, "p95": 0.0, "unit": "USD"}
  }
}
```

Instrument-stage template (p004):

You are performing a structured quantitative elicitation of the three INSTRUMENT-level parameters of a canonical Bayesian value-of-information model, for one specific evaluation of one deployment decision. The decision-level parameters (the prior p and the stakes B, K) are elicited separately, by another elicitation, and are NOT asked here: do not report them, and do not let the stakes of the decision move your estimate of the evaluation's quality or cost.

The model

A single agent faces a single binary decision: respond ($a=1$) or do not respond ($a=0$).
The world is in a hidden binary state θ . Convention: $\theta=1$ is the state in which responding is the correct action.

The agent can buy exactly one measurement from an instrument (here: one run of one evaluation). It produces a binary signal x , where $x=1$ suggests $\theta=1$.

The three parameters you elicit:

- s: sensitivity $P(x=1 \mid \theta=1)$ - the probability the evaluation flags the condition when it is truly present.
- t: specificity $P(x=0 \mid \theta=0)$ - the probability the evaluation stays silent when the condition is truly absent.
- C: the total cost, in USD, to build this evaluation from scratch (design, data or scene collection, hardware, engineering time) and run it once against the system under test.

The other three (the prior p, the benefit B of a correct response and the false-alarm cost K, all for the deployment decision) are elicited separately. All USD amounts are per single decision. Never use market sizes, annual totals, portfolio-wide or fleet-wide numbers.

Anchor scenarios (calibrate your scales against these agreed numbers)

Anchor A1. A plant reliability engineer decides whether to pull a critical gearbox for overhaul. $\theta=1$ = an incipient bearing fault is present. Instrument = one third-party vibration-analysis survey.

- s: 0.60 / 0.80 / 0.95
- t: 0.70 / 0.90 / 0.98
- C: 300 / 1,000 / 5,000 USD

Anchor A2. A primary-care physician decides whether to start a patient with acute sore throat on antibiotics. $\theta=1$ = group A streptococcal infection is present. Instrument = one rapid antigen strep test.

- s: 0.70 / 0.85 / 0.95
- t: 0.90 / 0.95 / 0.99
- C: 5 / 15 / 50 USD

Each triple is p5 / p50 / p95 of the elicitor's belief. Note the scales: A1 is an industrial survey costing thousands of USD, A2 an everyday test costing tens of USD. Place the evaluation below on a consistent scale relative to both.

Scenario to elicit

- Title: \$title
- Agent (the decision-maker): \$agent
- Decision (respond vs not): \$decision
- State definition: \$theta_definition
- Instrument (the evaluation, run once): \$instrument

Background facts about this evaluation

\$context

Use these facts. Your reasoning and your numbers must not contradict them; where they give a quantity (a cost, a rate, a scale), take it as given and reason from it. Where they are silent, reason from base rates and typical magnitudes as usual.

```

## Instructions

For each of the three parameters, in the order s, t, C:
1. First write a short paragraph of reasoning (2-4 sentences): what drives this quantity for THIS
   evaluation, referencing the background facts, base rates or typical magnitudes where you can.
2. Then commit to three percentiles p5, p50, p95. Read them as: p50 is your median estimate; you would
   be genuinely surprised if the true value fell outside [p5, p95]. They must be strictly increasing:
   p5 < p50 < p95.
3. State the unit ("probability" or "USD").

Remember for s, t: they describe THIS evaluation's ability to flag the state defined above, given that
the state is what it is; a free substitute for the evaluation does not belong here (it lives in the
prior, elicited separately). The median s must exceed 1 minus the median t (the evaluation must be
informative).
Remember for C: build-from-scratch plus one run, not the marginal cost of a repeat run.

## Output

Output strict JSON only. No markdown fences, no prose outside the JSON. Exactly this shape, with all
three parameters filled in:

{
  "parameters": {
    "s": {"reasoning": "...", "p5": 0.0, "p50": 0.0, "p95": 0.0, "unit": "probability"},
    "t": {"reasoning": "...", "p5": 0.0, "p50": 0.0, "p95": 0.0, "unit": "probability"},
    "C": {"reasoning": "...", "p5": 0.0, "p50": 0.0, "p95": 0.0, "unit": "USD"}
  }
}

```

Decision contexts (p004 protocol file):

```

decision_contexts:
  home manipulator: >
    The decision is whether to delay the launch of a VLA-based (vision-language-action)
    mobile manipulator into customers' homes and add mitigations, or launch as planned;
    the hazard is an unsafe behaviour mode of the deployed policy that would injure a
    person or cause serious property damage in ordinary home use. Closed-loop simulation
    studies of current VLA policies report 13-15% unsafe-episode rates for tabletop
    baselines and find that 36-56% of nominally successful household rollouts violate a
    safety requirement (SafeVLA-Bench, RoboCasa365), while real-home deployments of
    end-to-end policies have shown simulation success of 77% falling to 23% in six real
    homes (CMU / Meta AI, 2023). Injury-relevant contact is governed by the body-region
    force limits of ISO 10218:2025 (which absorbed ISO/TS 15066), with measured pain
    thresholds of about 122-311 N; the only fleet-level harm data for robots come from
    warehouses (OSHA logs: 7.9 serious injuries per 100 workers per year at robotic
    warehouses, 54% above non-robotic ones). No source states the fleet size, unit price
    or launch schedule of this product; take a first consumer launch of a limited number
    of units as the deployment, and price B and K for that one launch decision.
  AV AEB: >
    The decision is whether to delay the AEB (automatic emergency braking) release of a
    vehicle programme and rework it, or release as planned; the hazard is a failure mode
    in which the system does not brake for a crossing pedestrian (adult or child,
    including a child emerging from behind parked cars) at 20-60 km/h in daylight or
    darkness. Per insured vehicle year, pedestrian AEB cuts pedestrian crashes by 25-27%
    and pedestrian injury crashes by 29-30% (IIHS, 2022), with no measurable effect in
    unlit darkness, at speed limits of 50 mph or more, or while turning; NHTSA's FMVSS
    127 rule requires no contact in every run of four pedestrian scenarios at 10-73 km/h
    in daylight and darkness, and its Standing General Order forces crash reporting for
    ADAS-equipped vehicles. Rare-event statistics need large exposure (RAND: 275M
    failure-free miles to bound a fatality rate at the 2013 US level with 95% confidence).
    No source states the programme's production volume, price or release date; take one
    vehicle programme's release across its production run as the deployment decision,
    and price B and K for that one release decision.

```

350 C Limitations

- 351 • **Elicited, not measured.** Every input is the belief of one vendor's language models about a
352 published evaluation described to them. The contexts were fact-checked, but the literature
353 supplies almost no cost per run and no cross-rung false-negative rate for safety, so *s*, *t* and
354 *C* are priors: in our catalogue 102 of 249 safety-focused robotics evaluations report no
355 validity evidence, and none of the 55 closed-loop safety benchmarks reports a correlation
356 with a real outcome. No rung is anchored to a measured sensitivity or specificity, there is

no non-Claude elicitor, and nothing here is validated against human experts; Sec. 5 lists the cheapest anchors.

- **Round numbers.** 80% of the Sonnet and Opus p50s of s and 67% of those of t sit on a multiple of 0.05, and Sonnet answers $s = 0.55$ in 22 of its 75 instrument elicitation, so the elicited J takes few distinct values.
- **Prompt and elicitor dependence.** The home gate verdict moves with the model set and with the prompt that elicited p (Sec. 4); the staged prompt also lowered C . The home decision context quotes unsafe-episode and violation rates from closed-loop simulation benchmarks (App. B), evidence from the L4 class of instrument that the ladder prices. A second decision-context wording without those rates, with the same models, would bound how much of “always respond” follows from them; it has not been run.
- **Decision model.** The action is binary and the state qualitative, so choosing and verifying mitigations, the main use of an evaluation once mitigation is expected, has no value; the graded-response Gaussian model values every rung (App. F). The model has no trial count per rung, no sim-to-real bias or correlation, and no combined use (a simulated screen followed by a real confirmation); rungs are substitutes.
- **Point estimate.** The plug-in uses pooled medians. For one risk-neutral decision-maker uncertain about independent parameters the value is (1) at the parameters’ predictive means (last column of Table 3), which depend on the fitted tails: B and K rise 2.9 to 3.6 times and the AEB EVPI from 4.8M to 21.6M, so the EVPI bound on fleet telemetry fails there, while the AEB order by net value (L8 first, L9 last) and the home verdict (every rung outside the gate, $\pi^* = 0.139$; $C(L9) = 1.26M > EVPI = 1.11M$) hold.
- **Scope.** Two ladders; the AEB ladder lacks five rungs by design; home L8 and L9 are composites; the top rungs’ C is dominated by the exposure that rare-event rates need [59, 62]. Rank correlations over the 5 AEB rungs cannot separate a moderate from a strong trend.
- **Per-decision value and build-from-scratch cost.** No reuse count (App. G); one decision per ladder. The AEB ordering by net value cannot be overturned by any discount on the simulated rungs: even free, their EVSI (1.29M, 1.45M) falls short of the track test’s net value (2.26M); the efficiency orderings and the home fence ordering can (L4’s EVSI*, 237k, exceeds L7’s fence net value, 219k).
- **Statistics.** The Monte Carlo (MC) median is zero on every scenario (App. D); the bootstrap measures agreement between elicitation, not the width of each elicitation. The stored MC draws each rung’s p , B , K independently; only the step analysis shares draws along a ladder.
- **Gaussian check.** The state variable is chosen by the elicitor and differs between repeats on some rungs; the derived s , t disagree with the elicited ones.

D Net values, bootstrap intervals, Monte Carlo summaries and rung steps

Table 3 gives the elicited s and t , the net values and gate bands of the body, and the net values at the predictive means. Table 4 gives the secondary patterns of Sec. 4 in the format of Table 1. Table 5 gives the plug-in point with bootstrap intervals, Table 6 the bootstrap rank intervals and Table 7 the rung steps; Fig. 3 draws the plug-in point with its intervals. 6 of 15 scenarios are in the gate at the plug-in point.

Monte Carlo summaries. An MC run draws 100 000 times from the mixture of the fitted distributions and reports the mean, the median, $P_+ = P(EVSI > C)$ and $P_{\text{gate}} = P(EVSI > 0)$. The mixture spreads each rung across regimes: home rungs always respond on 47 to 58% of draws, and over all rungs P_+ is 3% to 39% and P_{gate} 15% to 45%, so the MC median EVSI is zero on all 15 scenarios and ranks nothing. The MC median has no decision-theoretic reading. The MC mean is

Table 3: Per rung at the plug-in point (staged prompt, Sonnet+Opus): pooled elicited medians s , t and C (USD), $J = s+t-1$, net value $\text{EVSI} - C$ and fence net value $\text{EVSI}^* - C$ (USD), and the gate band, the priors for which the rung can change the decision at this decision’s π^* (0.167 home, 0.184 AEB; the elicited p is 0.335 and 0.12). ΔJ and $\Delta J^* = \Delta C / (\Delta p(1-p))$: the step from the previous rung and its break-even gain for the fence buyer, whose step pays if $\Delta J > \Delta J^*$. Last column: net value at the parameters’ predictive means (the equal-weight mean of each valid elicitation’s fitted Beta or lognormal), where every home rung is outside the gate ($\pi^* = 0.139$, $\text{EVSI} = 0$). Full rung names: Table 2.

id	rung	s	t	J	C	$\text{EVSI} - C$	$\text{EVSI}^* - C$	gate band	ΔJ	ΔJ^*	at means
<i>home manipulator</i>											
1	L0 text prompts	0.35	0.75	0.100	25k	−25k	55k	[0.13, 0.19]			−37.3k
2	L1 image QA	0.66	0.525	0.185	35k	−35k	113k	[0.13, 0.24]	0.085	0.012	−45.7k
3	L2 video QA	0.55	0.725	0.275	90k	−90k	131k	[0.09, 0.24]	0.090	0.069	−130k
4	L3 adv. scenes	0.65	0.50	0.150	50k	−50k	70k	[0.13, 0.22]	−0.125	−0.050	−85.9k
5	L4 MuJoCo suite	0.665	0.63	0.295	42.5k	−42.5k	194k	[0.10, 0.27]	0.145	−0.009	−66.6k
6	L5 world model	0.55	0.79	0.340	165k	−165k	108k	[0.07, 0.26]	0.045	0.153	−382k
7	L6 HIL twin	0.55	0.79	0.340	200k	−200k	73k	[0.07, 0.26]	0.000	0.044	−357k
8	L7 lab trials	0.735	0.725	0.460	150k	−127k	219k	[0.07, 0.35]	0.120	−0.062	−170k
9	L8 field test	0.60	0.81	0.410	225k	−225k	104k	[0.06, 0.29]	−0.050	0.094	−291k
10	L9 pilot	0.55	0.82	0.370	600k	−600k	−303k	[0.06, 0.27]	−0.040	0.468	−1.26M
<i>AEB</i>											
11	L1 VLM QA	0.55	0.69	0.240	32.5k	152k	1.21M	[0.11, 0.26]			4.79M
12	L4 virtual test	0.50	0.86	0.360	500k	791k	1.36M	[0.06, 0.28]	0.120	0.090	6.04M
13	L6 vehicle-in-loop	0.55	0.85	0.400	600k	852k	1.47M	[0.06, 0.30]	0.040	0.019	7.62M
14	L8 track test	0.78	0.85	0.630	300k	2.26M	2.96M	[0.04, 0.47]	0.230	−0.058	12.0M
15	L9 fleet telemetry	0.50	0.87	0.370	3M	−1.63M	−1.09M	[0.06, 0.28]	−0.260	0.522	1.75M

Table 4: Secondary patterns, in the format of Table 1 (headline: staged prompt, Sonnet+Opus, plug-in point). ρ : Spearman correlation with level over the rungs of a ladder (home, AEB); with 5 AEB rungs its AEB values are descriptive. ρ' : Spearman correlation between two rankings of the 15 rungs. The first row follows from the range of C along a ladder (a factor of 24 on the home ladder and 92 on the AEB ladder, against 4.6 and 2.6 for J) rather than from the elicited s and t .

Pattern	Headline	Also holds under	Fails or weakens under
Fence efficiency EVSI^* / C against level	$\rho - 0.70$, both ladders	each model alone (home -0.64 to -0.56 , AEB -0.90 to -0.60); single-prompt plug-in (-0.72 , -0.60); Gaussian step efficiency (-0.71 , -0.90)	one model under a single prompt (Opus home -0.32 , Haiku AEB $+0.10$); gated home plug-in ($+0.29$)
Specificity, not sensitivity, tracks the level	t 0.66, 0.67; s 0.16, -0.05	single prompt (t 0.66, 0.60; s -0.01 , -0.21); all three models (t 0.73, 0.80; s 0.26, 0.10); 4 of 6 model-ladder pairs	Opus AEB (s $+0.67$, t $+0.30$); Sonnet home mixed (s $+0.46$, t $+0.74$)
Threshold-free ranking reproduces; gated ranking does not	ρ' bootstrap vs point: plug-in 0.93, fence 0.97	ρ' fence vs Gaussian fence-type 0.69 to 0.82 (per dollar 0.62 to 0.77); 0.96 to 0.97 on 68 decisions of a separate study	ρ' gated plug-in vs Gaussian plug-in (-0.15 to 0.06)

the value to an agent who learns the parameters before deciding, a different question from the plug-in’s; it agrees with the plug-in ranking at Spearman 0.63 but averages over a mass at zero. The fence ranking agrees with the plug-in where the plug-in is positive (Spearman 0.89, $n = 6$). $P_{\text{boot}}(\text{gate})$ (bootstrap of the medians) and P_{gate} (MC draws) answer different questions: the mixture pairs one repeat’s stakes with another’s prior and pulls every rung towards the middle. AEB L1 sits 52% in the gate over the bootstrap because the prior lies just above its lower gate edge (0.113).

Rung steps on common random numbers (CRN). Drawing p , B , K once per ladder and s , t , C per rung, a step pays with probability 15% (AEB L1→L4) and 4% (L8→L9); the CRN median of ΔEVSI is zero on every step, so Table 7 gives the plug-in steps only.

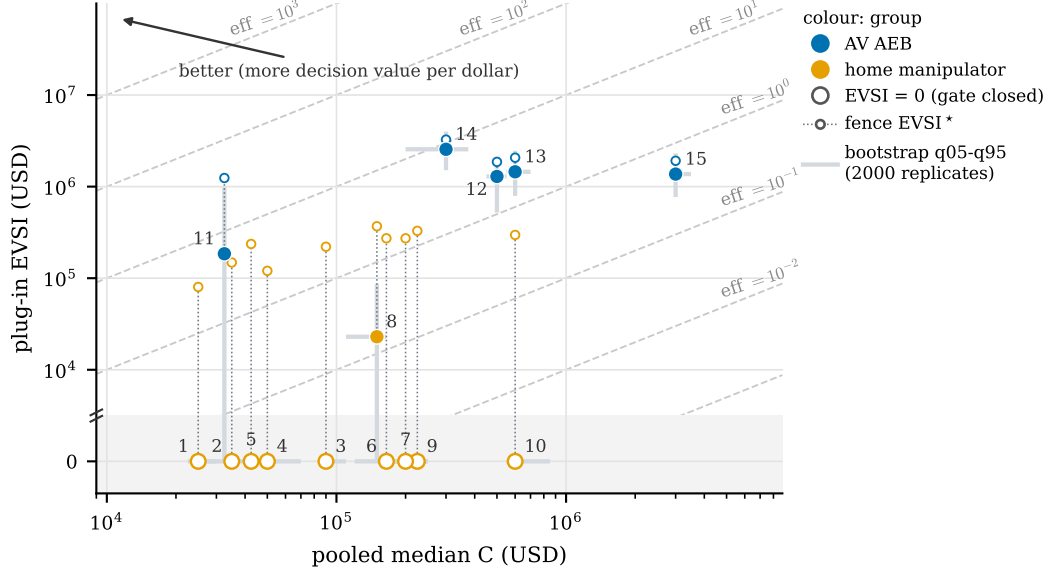


Figure 3: Plug-in EVSI against cost (staged prompt, Sonnet+Opus; “AV AEB”: the pedestrian AEB ladder). Filled: plug-in EVSI; open, on stems: fence value $EVSI^* = \Lambda p(1 - p)J$; on the floor: $EVSI = 0$, outside the gate. Grey bars: bootstrap q05–q95 on C and on EVSI. Dashed: constant efficiency. Labels are scenario ids (Table 2): home rung Lk is id $k+1$; AEB ids 11 to 15 are L1, L4, L6, L8, L9.

Table 5: Plug-in values per rung (staged prompt, Sonnet+Opus): Eq. (1) at the pooled medians of p , s , t , B , K and C , the point of Table 3. id: Table 2. $EVSI^* = \Lambda p(1 - p)J$: the fence value (Sec. 2); $eff = EVSI/C$ and $eff^* = EVSI^*/C$, each followed by its bootstrap [q05, q95]; $EVSI/EVSI^*$: the share of the fence value this decision-maker obtains (0 outside the gate); regime at the plug-in point: in the gate, or always respond (both posteriors above π^*); $P_{boot}(\text{gate})$: the share of bootstrap replicates in the gate. Bootstrap: 2000 replicates, each resampling every model’s valid elicitations with replacement (the decision stage once per decision) and recomputing the pooled medians. Rows are in the order of $P(EVSI > C)$ over the MC draws, since the MC median efficiency is zero on every rung (App. D); rank is that order.

rank	id	C	EVSI	$EVSI^*$	eff	[q05, q95]	eff^*	[q05, q95]	$EVSI/EVSI^*$	regime	$P_{boot}(\text{gate})$
1	14	300k	2.56M	3.26M	8.52	[4.7, 14]	10.9	[7.5, 17]	0.78	gate	100%
2	13	600k	1.45M	2.07M	2.42	[1.3, 4.2]	3.45	[2.6, 5.1]	0.70	gate	100%
3	11	32.5k	185k	1.24M	5.69	[0, 31]	38.2	[27, 59]	0.15	gate	52%
4	12	500k	1.29M	1.86M	2.58	[1, 4.3]	3.73	[2.6, 5.5]	0.69	gate	100%
5	8	150k	23k	369k	0.153	[0, 0.66]	2.46	[2.1, 3.7]	0.06	gate	74%
6	5	42.5k	0	237k	0	[0, 0]	5.57	[4.3, 7.3]	0.00	always	4%
7	2	35k	0	148k	0	[0, 0]	4.24	[3.2, 5.7]	0.00	always	0%
8	4	50k	0	120k	0	[0, 0]	2.41	[1.9, 3.9]	0.00	always	0%
9	1	25k	0	80.2k	0	[0, 0]	3.21	[2.2, 5]	0.00	always	0%
10	6	165k	0	273k	0	[0, 0]	1.65	[1.1, 2.7]	0.00	always	1%
11	9	225k	0	329k	0	[0, 0.00027]	1.46	[1.2, 2]	0.00	always	5%
12	3	90k	0	221k	0	[0, 0]	2.45	[1.8, 3]	0.00	always	0%
13	15	3M	1.37M	1.91M	0.457	[0.24, 0.77]	0.638	[0.49, 0.9]	0.72	gate	100%
14	7	200k	0	273k	0	[0, 0]	1.36	[1, 1.7]	0.00	always	0%
15	10	600k	0	297k	0	[0, 0]	0.495	[0.33, 0.58]	0.00	always	0%

Table 6: Rank stability of the plug-in and fence rankings under the bootstrap of Table 5. Per ranking (rank 1 the largest, ties averaged): the rank at the plug-in point, the median and [q05, q95] of the replicate rank, and the share of replicates in which the rung ranks in the top 3. Rows in plug-in order. AEB L8 (id 14) stays first by plug-in efficiency in 69% of replicates; the median Spearman correlation between a replicate’s ranking and the point ranking is 0.93 (plug-in) and 0.97 (fence).

id	plug-in eff				fence eff*			
	point	median	[q05, q95]	$P(\leq 3)$	point	median	[q05, q95]	$P(\leq 3)$
14	1	1	[1, 2]	1.00	2	2	[2, 2]	1.00
11	2	4	[1, 10.5]	0.47	1	1	[1, 1]	1.00
12	3	3	[2, 4]	0.77	5	6	[4, 9]	0.05
13	4	3	[2, 4]	0.75	6	6	[4, 9]	0.02
15	5	5	[4, 6]	0.00	14	14	[14, 15]	0.00
8	6	6	[4, 10.5]	0.01	8	8	[6, 10]	0.00
5	11	10.5	[10, 11]	0.00	3	3	[3, 4]	0.86
2	11	10.5	[10, 11]	0.00	4	4	[4, 7]	0.05
1	11	10.5	[10, 11]	0.00	7	7	[4, 10]	0.02
3	11	10.5	[10, 11]	0.00	9	10	[8, 11]	0.00
4	11	10.5	[10, 11]	0.00	10	8	[5, 10]	0.00
6	11	10.5	[10, 11]	0.00	11	11	[9, 13]	0.00
9	11	10.5	[8, 11]	0.00	12	12	[11, 13]	0.00
7	11	10.5	[10, 11]	0.00	13	13	[11.5, 13]	0.00
10	11	10.5	[10, 11]	0.00	15	15	[14, 15]	0.00

Table 7: Rung steps at the plug-in point (staged prompt, Sonnet+Opus; subscript pi: plug-in): ΔEVSI and ΔC between adjacent rungs of a ladder, their ratio with its bootstrap [q05, q95], $P_{\text{boot}}(\text{pays})$, the share of replicates with $\Delta\text{EVSI} > \Delta C$, and the fence step ΔEVSI^* and its ratio. For $\Delta C < 0$ a step pays whenever $\Delta\text{EVSI} > \Delta C$, as AEB L6→L8 does. †: ΔC_{pi} takes each sign in at least 5% of the replicates, so the ratio straddles its pole.

step	ids	$\Delta\text{EVSI}_{\text{pi}}$	ΔC_{pi}	$\Delta\text{EVSI}_{\text{pi}}/\Delta C_{\text{pi}}$	[q05, q95]	P_{boot} (pays)	ΔEVSI^*	$\Delta\text{EVSI}^*/\Delta C_{\text{pi}}$
<i>pedestrian AEB</i>								
1→4	11→12	1.11M	468k	2.37	[0.88, 3.3]	94%	621k	1.33
4→6	12→13	161k	100k	1.61	[−0.67, 8.7]	73%	207k	2.07
6→8	13→14	1.1M	−300k	−3.68	[−5.9, −1.8]	100%	1.19M	−3.97
8→9	14→15	−1.19M	2.7M	−0.439	[−0.63, −0.25]	0%	−1.35M	−0.498
<i>home manipulator</i>								
0→1	1→2	0	10k	0	[0, 0]	2%	68.2k	6.82
1→2	2→3	0	55k	0	[0, 0]	0%	72.2k	1.31
2→3	3→4	0	−40k	0	[0, 0]	100%	−100k	2.51
3→4	4→5	0	−7.5k	0	[0, 0] [†]	64%	116k	−15.5
4→5	5→6	0	122k	0	[0, 0]	0%	36.1k	0.295
5→6	6→7	0	35k	0	[0, 0] [†]	15%	0	0
6→7	7→8	23k	−50k	−0.459	[−1.9, 0]	100%	96.2k	−1.92
7→8	8→9	−23k	75k	−0.306	[−1.5, 0]	0%	−40.1k	−0.535
8→9	9→10	0	375k	0	[−0.00016, 0]	0%	−32.1k	−0.0855

E Protocol iterations and elicitation health

p001 to p002. The log-decade framing of the dollar questions reduced the cross-repeat spread of B and K only modestly over all 5 repeats (B 4.40 to 4.00) and raised that of C ; over the first 3 repeats all three fell (B 2.50 to 1.40, C 1.10 to 1.00). It reordered the scenarios (Spearman correlation of median efficiency with p001 −0.17) and was reverted. **p003.** Adding Sonnet and Opus cut the dollar spread several-fold: under p003 the spread of B is 5.98 for Haiku, 1.00 for Sonnet and 0.67

for Opus. Haiku’s dollar answers stay far noisier and, under p004, its stakes are inverted ($K > B$), so the headline pools Sonnet and Opus only. The single prompt still let the instrument text move p , B and K (Table 9), and under p003 every rung sits close to the fence: all 15 scenarios are in the gate at the plug-in point while the MC median is zero on 14. **p004.** Staging fixes p , B , K per decision. Model-matched against p003, it lowered C for every model, by a median 0.13 (Opus) to 0.52 (Haiku) decades per rung, and moved the home prior from 0.15–0.26 to 0.335; the shift in s and t is small: Sonnet’s s moves by a median -0.03 per rung (sign test $P = 0.022$), Haiku’s by -0.03 (borderline, $P = 0.065$), and Opus’s shifts have no consistent sign (8 rungs up, 6 down). **MC-median rankings.** Of the 14 MC-median rankings (per protocol and model subset), five are constant: p003 (all three models), p003 Opus, p004 (all three, 15 of 15 medians zero), p004 Sonnet+Opus (15 of 15) and p004 Opus. p001 (11 zero) and p003 Sonnet+Opus (14) are nearly so. Hence 55 of the 91 protocol pairs have no defined rank correlation.

Health. Table 10 gives the elicitations this paper uses. Spread is the cross-repeat spread of a parameter, ($\max - \min$) over the pooled p50 of one scenario’s repeats, median over scenarios (over the two decisions for p , B , K under p004). Under p004, Haiku’s decision-level B for the AEB release runs from 1.5M to 900M USD across 5 repeats, which is why the headline pools Sonnet and Opus only. The Gaussian protocol lost attempts to a usage-limit outage of the model client: over all models 223 of 279 attempts were valid (USD 41.83); these counts, and the study total of App. H, include the outage rows, which were stored as invalid and re-run, while Table 10 leaves them out.

Table 8: Protocols. Models: Claude Haiku (H), Sonnet (S), Opus (O), 5 repeats each.

protocol	models	change against the previous	outcome
p001	H	baseline: one prompt per rung, all six parameters	decision numbers drift along a ladder
p002	H	log-decade framing of B , K , C	over all 5 repeats B , K spread slightly lower, C higher (first 3: all lower); re-ordered; reverted
p003	H, S, O	p001 template, three models	S and O far less noisy in dollars; drift remains
p004	H, S, O	staged: p , B , K once per decision; s , t , C per rung	headline (S+O); drift zero by construction
g001	H, S, O	Gaussian-state family (App. F)	model check

Table 9: The consistency check per protocol: dispersion of the pooled p50 across the rungs of each ladder (CV for probabilities, $\log_{10}$ range for dollars); the ratio column divides $\text{CV}(p)$ by the mean CV of s and t ; last column: p , B , K flatter than s , t , C . Top to bottom: p001 Haiku, p003 Sonnet+Opus, p004 Sonnet+Opus (headline).

group	levels	CV p	CV s	CV t	$\log_{10}$ range B	$\log_{10}$ range K	$\log_{10}$ range C	$\frac{\text{CV}(p)}{\text{CV}(s,t)}$	p , B , K flatter
pedestrian AEB	5	0.14	0.08	0.04	2.40	1.82	1.90	2.35	no
home manipulator	10	0.31	0.06	0.06	0.70	0.70	0.93	5.10	no
group	levels	CV p	CV s	CV t	$\log_{10}$ range B	$\log_{10}$ range K	$\log_{10}$ range C	$\frac{\text{CV}(p)}{\text{CV}(s,t)}$	p , B , K flatter
pedestrian AEB	5	0.11	0.20	0.10	0.57	0.57	1.63	0.74	no
home manipulator	10	0.17	0.15	0.18	0.20	0.03	1.46	1.00	no
group	levels	CV p	CV s	CV t	$\log_{10}$ range B	$\log_{10}$ range K	$\log_{10}$ range C	$\frac{\text{CV}(p)}{\text{CV}(s,t)}$	p , B , K flatter
pedestrian AEB	5	0.00	0.18	0.08	0.00	0.00	1.97	0.00	yes
home manipulator	10	0.00	0.17	0.16	0.00	0.00	1.38	0.00	yes

Table 10: Elicitation health of the protocols and model sets used here. Attempts include retries and exclude usage-limit outage rows; valid is the share of attempts that validated.

protocol, models	attempts	valid	USD	fit warn.	cross-repeat spread p, s, t, B, K, C						
p001, Haiku	78	96.2%	5.87	25	0.68	0.15	0.09	4.40	4.33	1.68	
p003, Sonnet+Opus	157	94.9%	10.68	0	1.08	0.42	0.22	1.47	2.00	1.08	
p004, Sonnet+Opus	170	100.0%	9.39	1	1.45	0.38	0.19	1.82	2.29	1.18	
p004, all three	255	100.0%	13.77	6	1.75	0.50	0.25	13.19	3.65	2.14	
g001, Sonnet+Opus	150	100.0%	31.06	48	Gaussian parameters, App. F						
<i>per model, p004 (headline)</i>											
Sonnet	85	100.0%	5.34								
Opus	85	100.0%	4.05								
<i>per model, p003</i>											
Sonnet	79	94.9%	5.91								
Opus	78	94.9%	4.77								

439 F The Gaussian-state family as a sanity check

440 **What it asks.** The Gaussian protocol (g001) asks the elicitor to name a continuous state with a
441 unit, its prior percentiles, the value θ_c at which the response changes, three routes to the sensor error
442 relative to the prior spread x , four loss points, the stakes and C , under a linear-Gaussian sensor
443 model. One reading removes a share $R^2 = 1/(1+x^2)$ of the prior variance of the state (the squared
444 correlation between reading and state). Four action models are scored from the same answers:
445 graded response with loss exponent k (quad), binary action with linear payoff (lin), step payoff with
446 a continuous reading (step), and step payoff with a pass/fail report at a fixed mark (stepfix), which
447 maps one-to-one onto the binary model and alone has a gate. Notation: $d = (\mu_0 - \theta_c)/\sigma_0$ is the
448 standardised distance of the prior mean μ_0 from θ_c (σ_0 the prior standard deviation), Φ the standard
449 normal distribution function, so the implied prior is $p = \Phi(d)$, and L the loss scale, the EVPI of the
450 graded-response model, whose value is LR^2 for $k = 2$. The stepfix mark sits at θ_c .

451 **What it confirms.** The direction of the step efficiency: the MC median of eff_{step} correlates with
452 level at -0.71 (home) and -0.90 (AEB). The threshold-free rankings: the binary fence value agrees
453 with the Gaussian fence-type values at 0.69 to 0.82 under p004 (Table 11, Fig. 4) and at 0.71 to
454 0.88 under p003 (the same comparison against the single-prompt run, not tabulated), and at 0.96 to
455 0.97 over 68 decisions of a separate study from other domains, elicited the same way (Sonnet+Opus;
456 binary plug-in against stepfix 0.74 there).

457 **The home gate.** The Gaussian run’s implied home prior $\Phi(d)$ is 0.16 to 0.28, below the upper gate
458 edges of the binary run, and the stepfix value is positive on every rung. At the Gaussian plug-in
459 point stepfix and step both exceed cost on the same 8 of 10 home rungs; over the MC draws the
460 median exceeds cost on 3 rungs for stepfix and 7 for step. Within this family the preset mark lowers
461 the value of a reading but does not close a gate at these priors, so the family cannot show how much
462 of the binary verdict is the mark.

463 **Where it disagrees.** The gated plug-in: -0.15 to 0.06 against the staged binary run, because the
464 binary plug-in is zero on the 9 always-respond home rungs while every Gaussian model is positive;
465 against the single-prompt run, where every rung is in the gate, it is 0.51 to 0.72 . The implied
466 instrument: the derived s (0.70 to 0.79) and t (0.86 to 0.94) are nearly flat across rungs and sit above
467 the elicited ones (mean difference $+0.166$ and $+0.151$; Spearman 0.32 and 0.42). The derived prior
468 has the smallest mean difference (-0.087), but its median absolute difference (0.13) exceeds that
469 of t (0.09), and because the elicited p takes one value per ladder, its Spearman $\rho = 0.75$ mostly
470 measures the separation of the two ladders (Fig. 5). R^2 sits at 0.60 to 0.78 on every rung and falls
471 with the rung on the home ladder (Spearman -0.81), so the Gaussian ranking is set by the loss scale
472 over cost, as the binary one is set by cost. **Caveats.** The elicitor chooses the state variable, and on
473 7 rungs the repeats pool physically different states (a rate with a contact force or an impact speed);
474 109 parameter fits in 58 elicitations carry a fit warning over all models (48 for Sonnet+Opus).

Table 11: Binary against Gaussian rankings over the 15 rungs, each family at its pooled medians (headline binary elicitations against the Gaussian ones, both Sonnet+Opus). Top: the gated binary plug-in efficiency against the four Gaussian action models. Bottom: the binary fence value $\text{EVSI}^* = \Delta p(1 - p)J$ against three threshold-free Gaussian values (stepfix at $\Phi(d) = p$, step at $\Phi(d) = \pi^*$, and the quad value LR^2 with $k = 2$), as values and per dollar; no gate applies to these rows.

binary	Gaussian	Spearman ρ	Kendall τ	top-10 overlap
eff	eff _{quad}	0.06	0.06	8 / 10
eff	eff _{jin}	-0.11	-0.08	7 / 10
eff	eff _{step}	-0.11	-0.08	8 / 10
eff	eff _{stepfix}	-0.15	-0.11	8 / 10
EVSI*	EVSI _{stepfix}	0.69	0.50	9 / 10
eff*	EVSI _{stepfix} /C	0.62	0.49	9 / 10
EVSI*	EVSI _{step} at $\Phi(d) = \pi^*$	0.75	0.59	9 / 10
eff*	EVSI _{step} /C	0.66	0.50	9 / 10
EVSI*	LR^2	0.82	0.69	9 / 10
eff*	LR^2/C	0.77	0.68	9 / 10

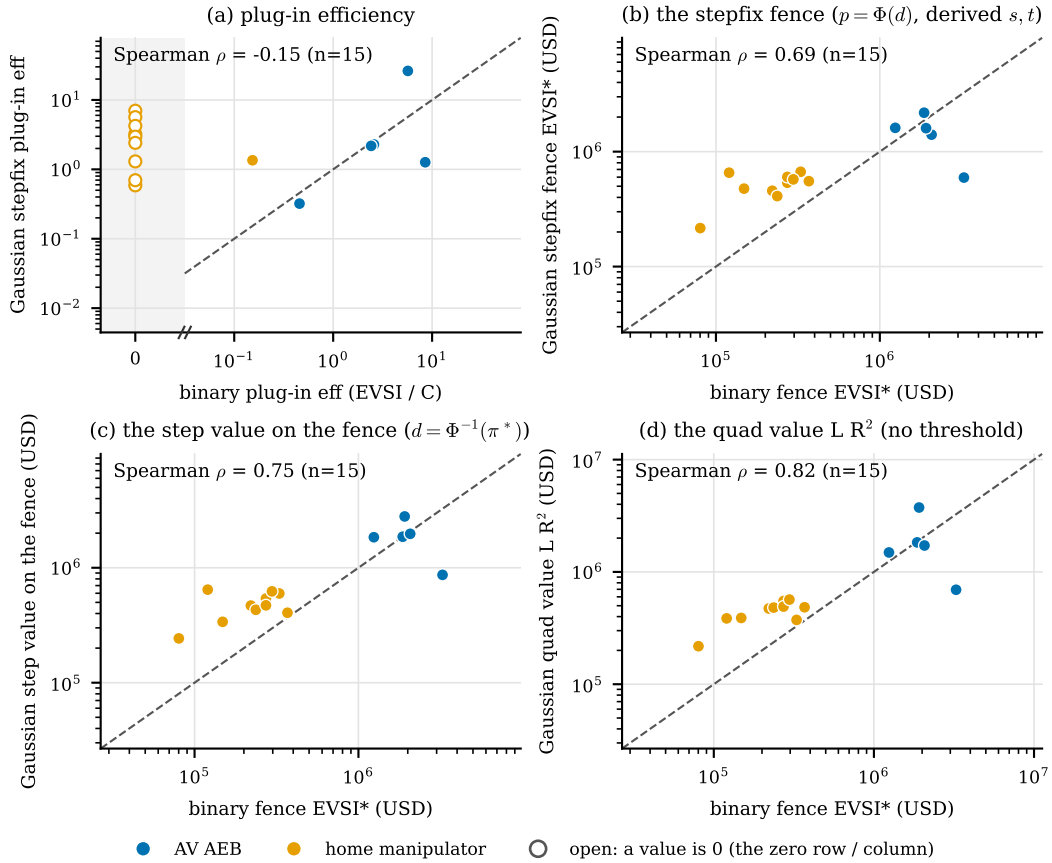


Figure 4: Binary against Gaussian plug-in points (headline binary elicitations against the Gaussian ones; “AV AEB”: the pedestrian AEB ladder): (a) gated efficiencies, binary plug-in against Gaussian stepfix; (b)–(d) the binary fence value $\text{EVSI}^* = \Delta p(1 - p)J$ against three Gaussian fence-type values (Table 11). Dashed: $y = x$.

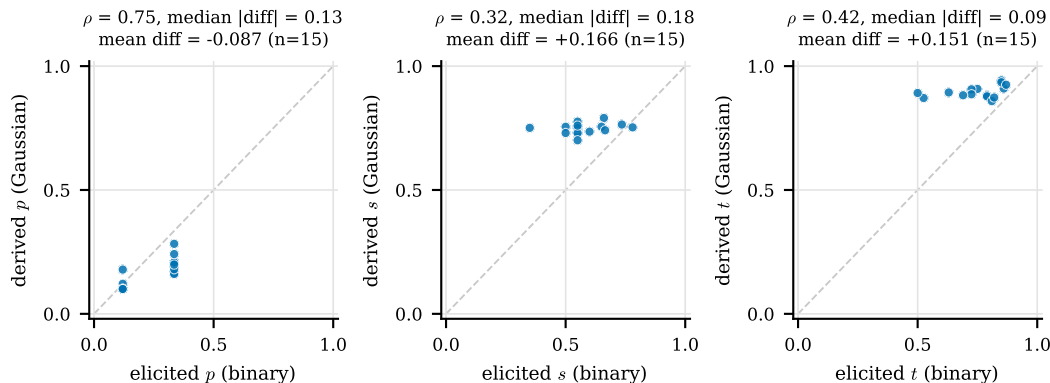


Figure 5: Derived $p = \Phi(d)$, s and t of the Gaussian elicitations (stepfix, mark at θ_c) against the elicited binary p50 (Sonnet+Opus). Dashed: $y = x$.

G Model choice

Why the efficacy multiplier e was dropped. An earlier version multiplied B by e , the probability that the agent acts on the signal and that the action works. It conflated compliance (part of the decision rule) with efficacy (part of B), discounted the benefit branch but not the false-alarm branch, and barely mattered: over the 17 top-quartile decisions of a separate study its mean $|\rho|$ against efficiency was 0.05 against 0.40 for C , and dropping it left the ranking of all 68 decisions in place (Spearman 0.996).

Why s and t stay. They create the gate: every zero in Fig. 2 is theirs. Dropping them collapses the ranking to $EVPI/C$; at the headline plug-in point that ranking agrees with $EVSI/C$ at 0.89 inside the gate but only 0.44 over all rungs. Along a ladder they are, with C , the only inputs that differ by design; the fence value is linear in Youden’s index and, inside the gate, the decision’s own value is linear in s and t (2).

What to add next: the reuse count. With C defined as build plus one run, a simulation suite built once and run on every release is charged its whole build cost for each decision. For an evaluation that informs n decisions the net value is $n EVSI - C_{\text{build}} - n C_{\text{run}}$; $n = 1$ reproduces the present objective. C_{build} , C_{run} and n belong to the instrument stage of the staged protocol, next to s and t . Per decision a rung can gain at most its C from reuse, so on the AEB ladder, where the simulated rungs’ $EVSI$ is below the track test’s net value, no reuse count changes the first place; on the efficiency orderings and on the home fence ordering it could, and eliciting n would tell.

H Reproduction

Every figure, every generated table and every generated number in this paper is regenerated from the stored elicitation database by one script, with no model calls, and the few numbers no generator writes are defined in one file, each next to the query or formula that produced it; the MC runs are deterministic given their seeds (seed 42, 100 000 draws). Stored runs and the hash of the valid elicitations each draws from: run 12 (headline: p004, Sonnet+Opus) 4a593132f091; run 11 (p004, all three models) fbd86ba76a3e; run 5 (p003, Sonnet+Opus) 06f845970ed1; run 1 (p001, Haiku) 2275676730b1; run 6 (g001, Sonnet+Opus) 0bbeb2e7455c. The model client was version 2.1.280 (Claude Code); model ids are in App. B. The whole study cost 83.40 USD over 925 elicitation attempts, outage rows included. Every attempt is stored with its prompt hash and raw response; scenarios, templates, protocol files, the literature catalogue and the fact-check log will be released with the code.